MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 27: Examples of linear-fractional conformal maps

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Lecture outline

1. Brief review of linear-fractional function
2. Examples of linear-fractional conformal maps

Linear-fractional map: elementary transformations

Linear fractional function is a fraction of two linear functions:

$$w = \frac{az+b}{cz+d} = \frac{a}{c} + \frac{bc-ad}{c(az+b)} = \alpha + \frac{\beta}{z+\gamma},$$

where $c \neq 0$, $ad - bc \neq 0$. If $ad - bc = 0$, $w(z)$ is a constant map, and if $c = 0, d \neq 0$, w is a linear map. Linear fractional maps can be considered as a sequence of the following elementary maps:

1. $w_1 = z + \gamma$ - parallel transfer
2. $w_2 = 1/w_1$ - inversion
3. $w_3 = |\beta| w_2$ - scaling
4. $w_4 = e^{i \arg \beta} w_3$ - rotation
5. $w_5 = w_4 + \alpha$ - parallel transfer

Linear-fractional map: basic properties

Some important properties of a linear-fractional function $w(z)$ are

- $w(z)$ is a 1-1 conformal map of an extended complex plane
- $w(z)$ maps circles (in extended sense including straight lines) onto circles
- $w(z)$ maps symmetrical with respect to a circle points into points that are symmetrical with respect to the image of this circle (symmetrical to the circle points means that all circles passing through these points are orthogonal to the circle).
- There is only one $w(z)$ mapping 3 different points a, b, c into 3 different points A, B, C of the extended plane

$$\frac{w-A}{w-B}\frac{C-B}{C-A}=\frac{z-a}{z-b}\frac{c-b}{c-a}$$

  If one of these points is ∞ then the formula correct with some adjustments, for example if $c=\infty$ then $\frac{c-b}{c-a}$ is equal to 1.

Example 1. Part 1.

Find a general linear-fractional function $w(z)$ mapping upper half-plane onto unit disk centered at the origin.

Let us consider a linear-fractional map in the following form:

$$w = k\frac{z-a}{z-b},$$

where k, a, b are unknown complex numbers.

There should be a point on a $z-plane$, let us call it α, which will be mapped into the origin $w(\alpha) = 0$. Then, the point symmetrical to α with respect to the real axis will be mapped to ∞, $w(\bar{\alpha}) = \infty$. Therefore, $a = \alpha$, $b = \bar{\alpha}$, and

$$w = k\frac{z-\alpha}{z-\bar{\alpha}},$$

In order to find k let us consider where $w(z)$ the boundary points. $w(z)$ must map boundary onto boundary, and therefore points of the real axis $z = x$ must map onto the circle $w(z) = 1$.

Example 1. Part 2.

Then

$$|w| = \left| k \frac{z-\alpha}{z-\bar{\alpha}} \right| = |k| \frac{|z-\alpha|}{|\overline{z-\alpha}|}$$

Observe that $|z-\alpha| = |\overline{z-\alpha}|$ as these numbers are complex conjugates. Therefore, $|k| = 1$, or $k = e^{i\theta}$, where θ is a real number.

Finally, the general form of a linear-fractional function mapping the upper half-plane onto the unit disk with the center at the origin is

$$w = e^{i\theta} \frac{z-\alpha}{z-\bar{\alpha}}, \tag{1}$$

where $Im\,\alpha > 0$ and β is a real number.

Example 2. Part 1.

Map upper half-plane onto unit disk $|w| < 1$ such that $w(2i) = 0$ and $arg\ w'(2i) = 0$.

Using formula 1 from previous slide and the given condition $w(2i) = 0$, we have $\alpha = 2i$ and $\bar{\alpha} = -2i$ and

$$w = e^{i\theta} \frac{z - 2i}{z + 2i}.$$

In order to find θ we first compute derivative of w at $2i$:

$$w'(z) = e^{i\theta} \frac{4i}{(z + 2i)^2},$$

$$w'(2i) = e^{i\theta} \frac{4i}{(4i)^2} = e^{i\theta} \frac{-i}{4}.$$

We rewrite this number in the exponential form:

$$e^{i\theta} \frac{-i}{4} = e^{i(\theta - \pi/2)}$$

Example 2. Part 2.

We apply condition $arg\, w'(2i) = 0$ to find θ:

$$\theta - \frac{\pi}{2} = 0$$

$$\theta = \frac{\pi}{2}$$

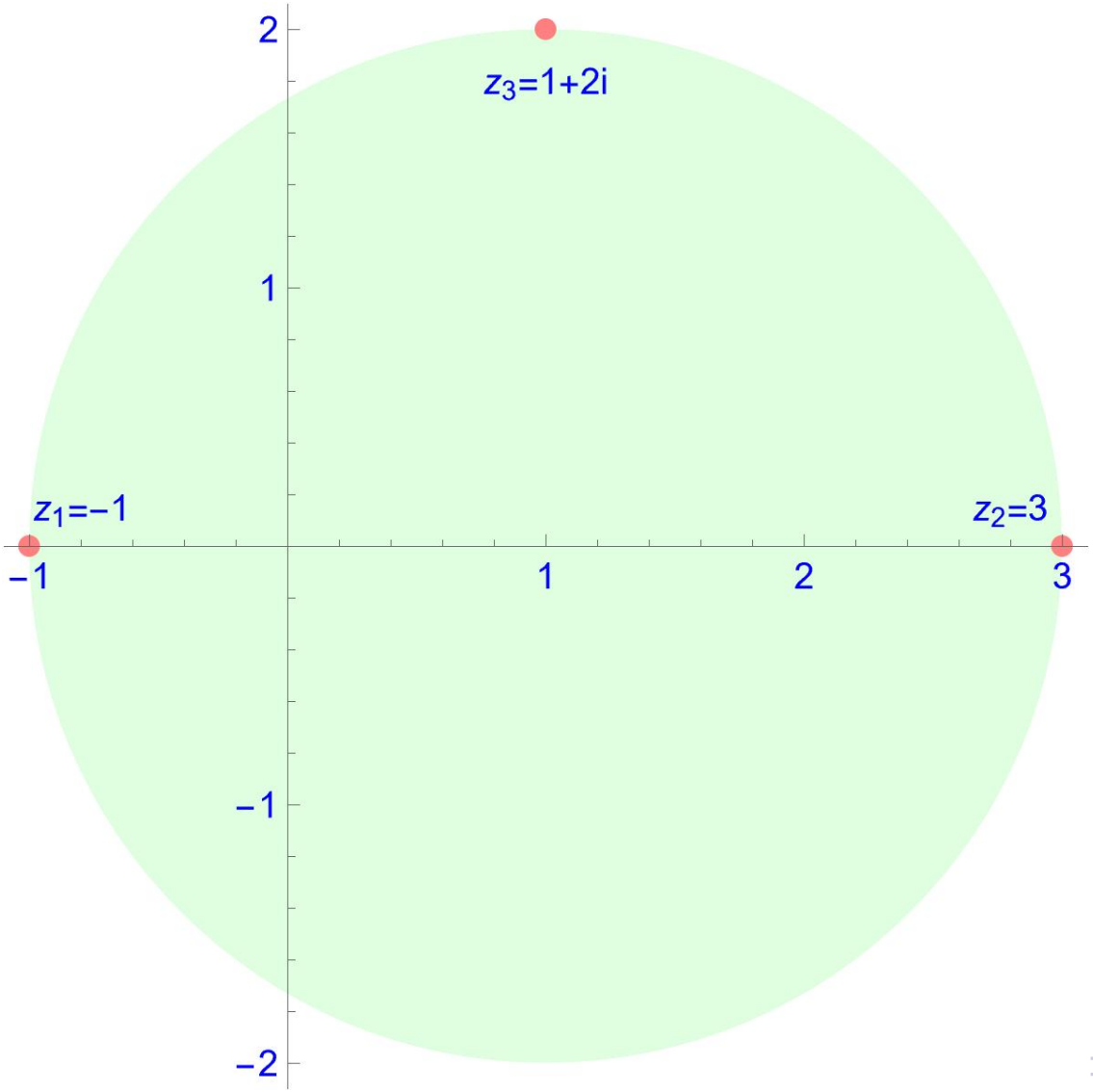
Finally, the following function maps upper half-plane $Im\, z > 0$ onto unit disk $|w| < 1$ under conditions $w(2i) = 0$ and $arg\, w'(2i) = 0$:

$$w = e^{i\pi/2}\frac{z-2i}{z+2i} = i\frac{z-2i}{z+2i} = \frac{iz+2}{z+2i}$$

Please note that this function is unique. To understand it please note that the general formula 1 include two unknown parameters and the problem formulation specified two initial conditions.

Example 3. Part 1.

Find image of the set $E = \{z : |z-1| < 2\}$ under the map $w = \dfrac{2iz}{z+3}$



Example 3. Part 2.

Let us pick 3 points on the circle $|z-1|=2$: $z_1=-1$, $z_2=3$, $z_3=1+2i$.

First, we notice that none of the points on the circle $|z-1|=2$ is mapped to ∞. In fact, $w(z)=\infty$ is possible only for $z=-3$, which is not on this circle.

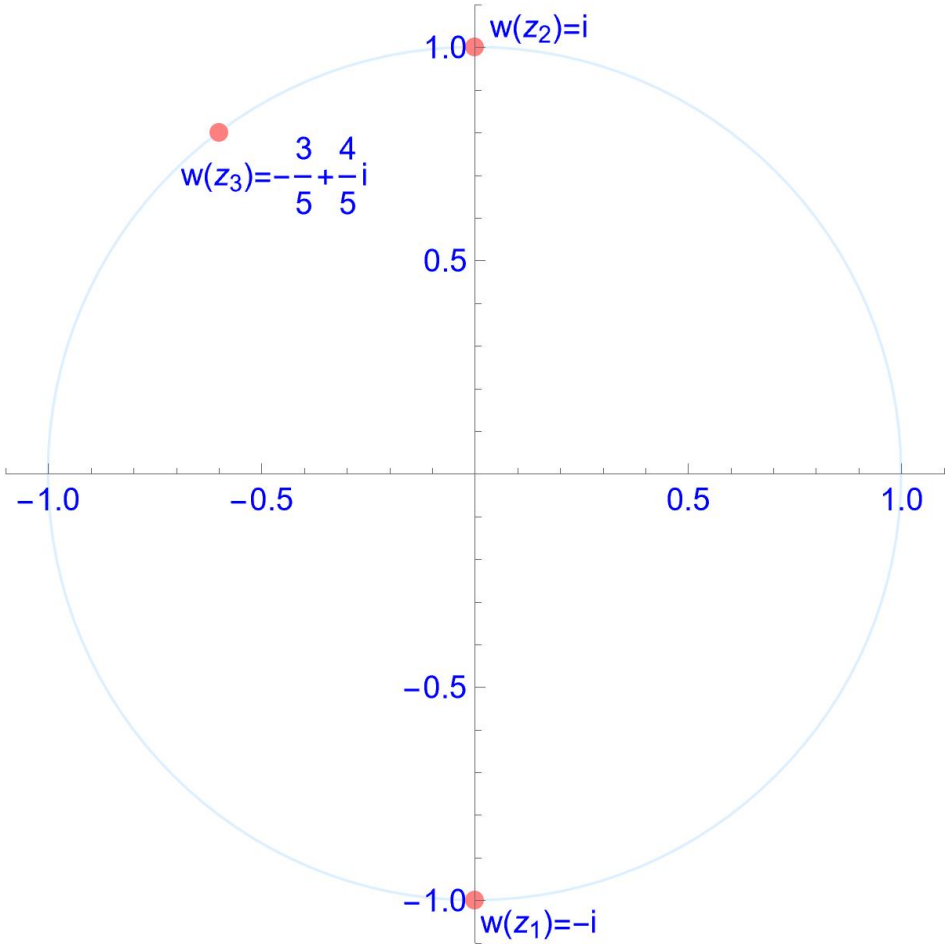
Therefore, the image of this circle is the circle (not a straight line!). The image circle should pass through images of these points:

$$\begin{aligned} w(z_1) &= w(-1) = -i \\ w(z_2) &= w(3) = i \\ w(z_3) &= w(1+2i) = -\frac{3}{5} + \frac{4}{5}i \end{aligned}$$

In this case, it is easy to visualize and guess the image circle. In general, we can resolve equation of the circle analytically using coordinates of these 3 points.

Example 3. Part 3.

Therefore, the circle $|w| = 1$ is the image of the boundary of the disc $E = \{z : |z - 1| < 2\}$ under the given map $w(z)$.



Example 3. Part 4.

Now, we need to find out where $w(z)$ maps the interior of the disk $E = \{z : |z-1| < 2\}$. This image can be interior or exterior of the circle $|w| = 1$.

We can use two different approaches to answer this question:

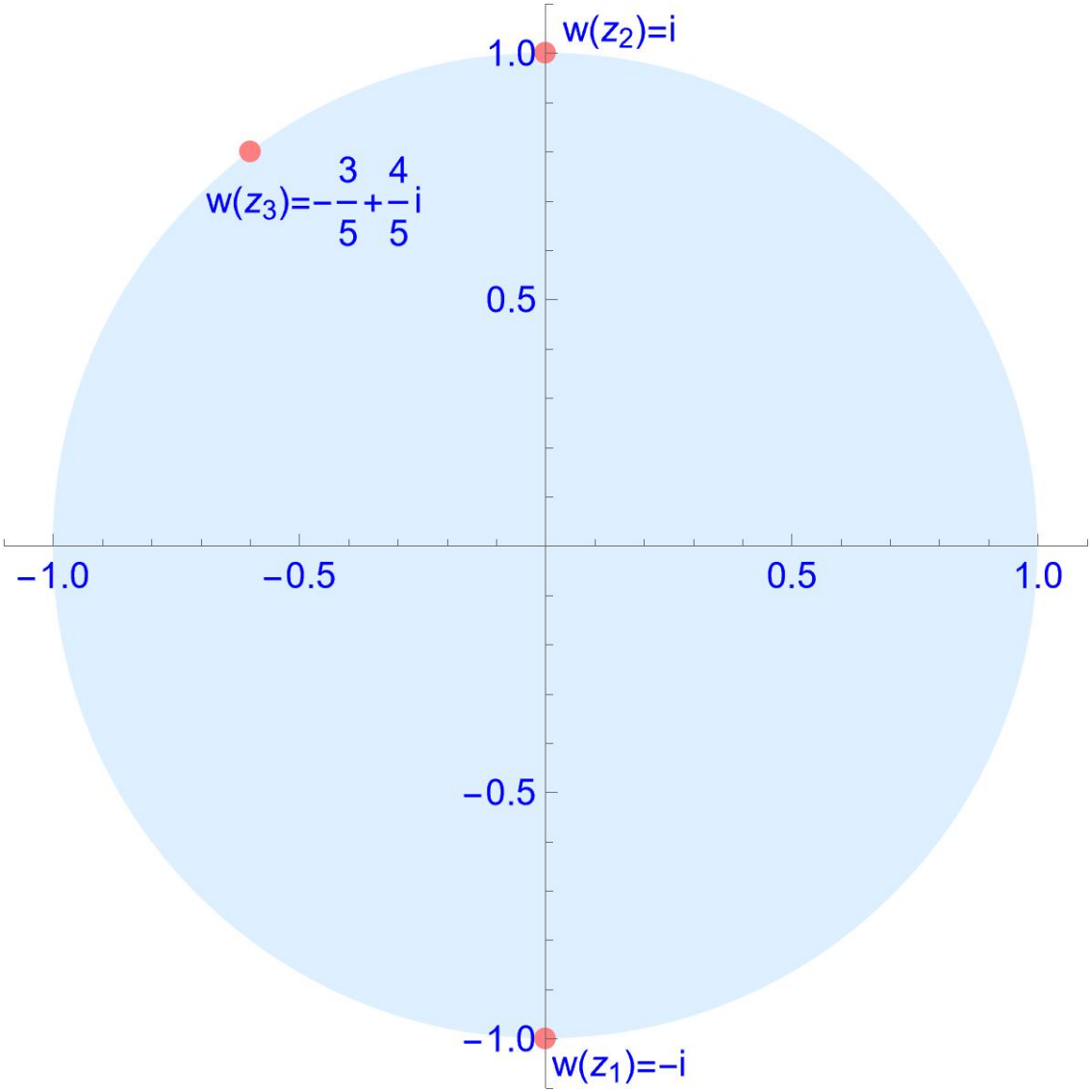
1. We can pick up any point of the interior (or exterior) or the circle $|z-1| = 2$ and check if its image is in the interior (or exterior) of the circle $|w| = 1$.
2. We can go around the circle $|z-1| = 2$ in the positive (negative) direction and check if the image point $w(z)$ moves around the circle $|w| = 1$ in the same or opposite positive (negative) direction.

First approach. Let us pick point $z = 0$ from the interior of the circle $|z-1| = 2$. Its image $w(0) = 0$ is in the interior of the circle $|w| = 1$. Therefore, $w(z)$ maps disk $E = \{z : |z-1| < 2\}$ onto disc $|w| < 1$.

Second approach. If we walk about the circle $|z-1| = 2$ in the positive direction starting from z_1, we pass our 3 points in the order: $z_1 = -1$, $z_2 = 3$, and $z_3 = 1 + 2i$. The disc interior is to the left of us. Now, we move around the image circle $|w| = 1$ in the positive direction from the point $w(z_1)$: the order of image points we pass is the same $w(z_1)$, $w(z_2)$, and $w(z_3)$ and the interior of $|w| = 1$ will be to the left. Therefore, $w(z)$ maps disk $E = \{z : |z-1| < 2\}$ onto disc $|w| < 1$.

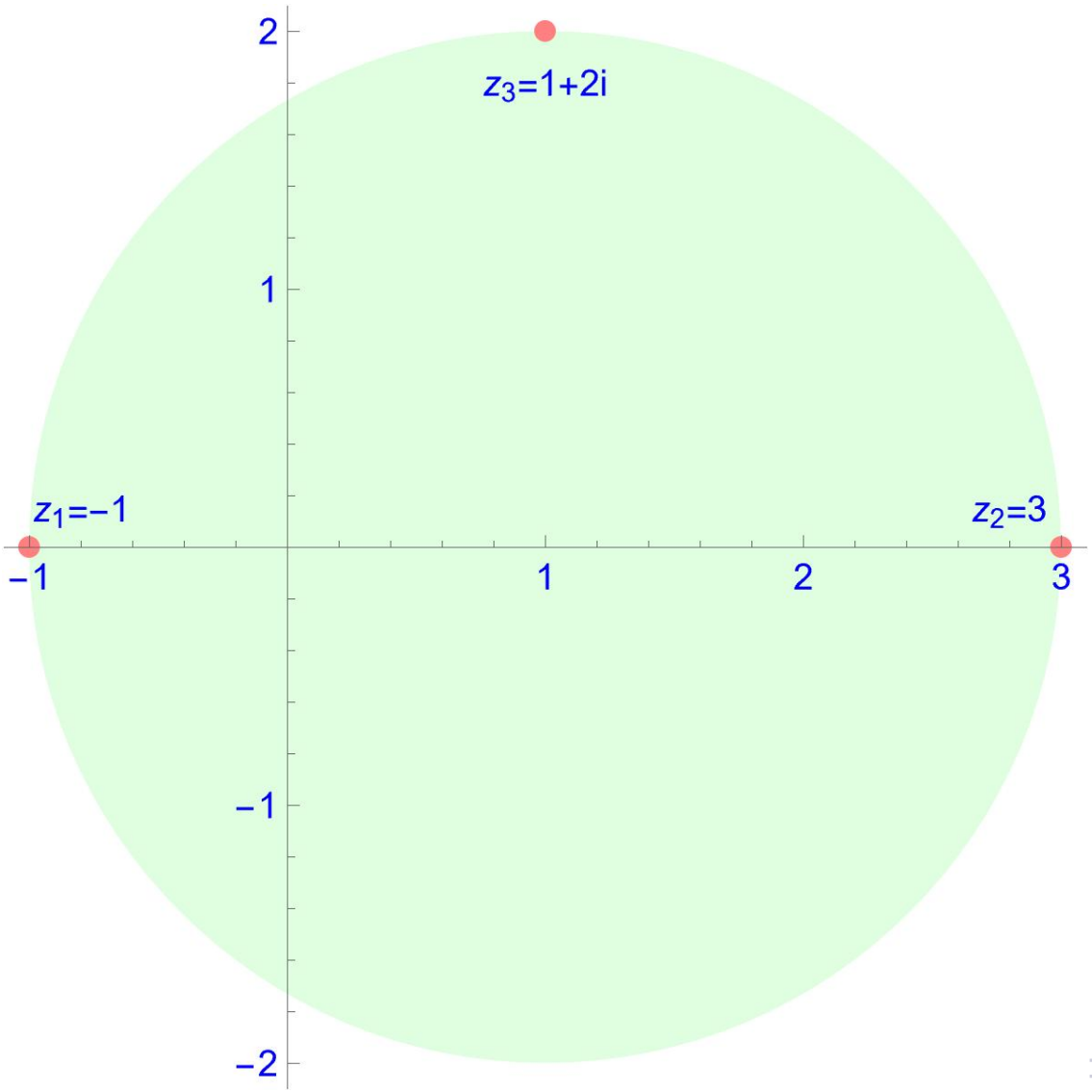
Example 3. Part 5.

Our final answer: $w(z)$ maps disk $E = \{z : |z-1| < 2\}$ onto disc $|w| < 1$.



Example 4. Part 1.

Find image of the set $E = \{z : |z-1| < 2\}$ under the map $w = \dfrac{z+1}{z-2}$



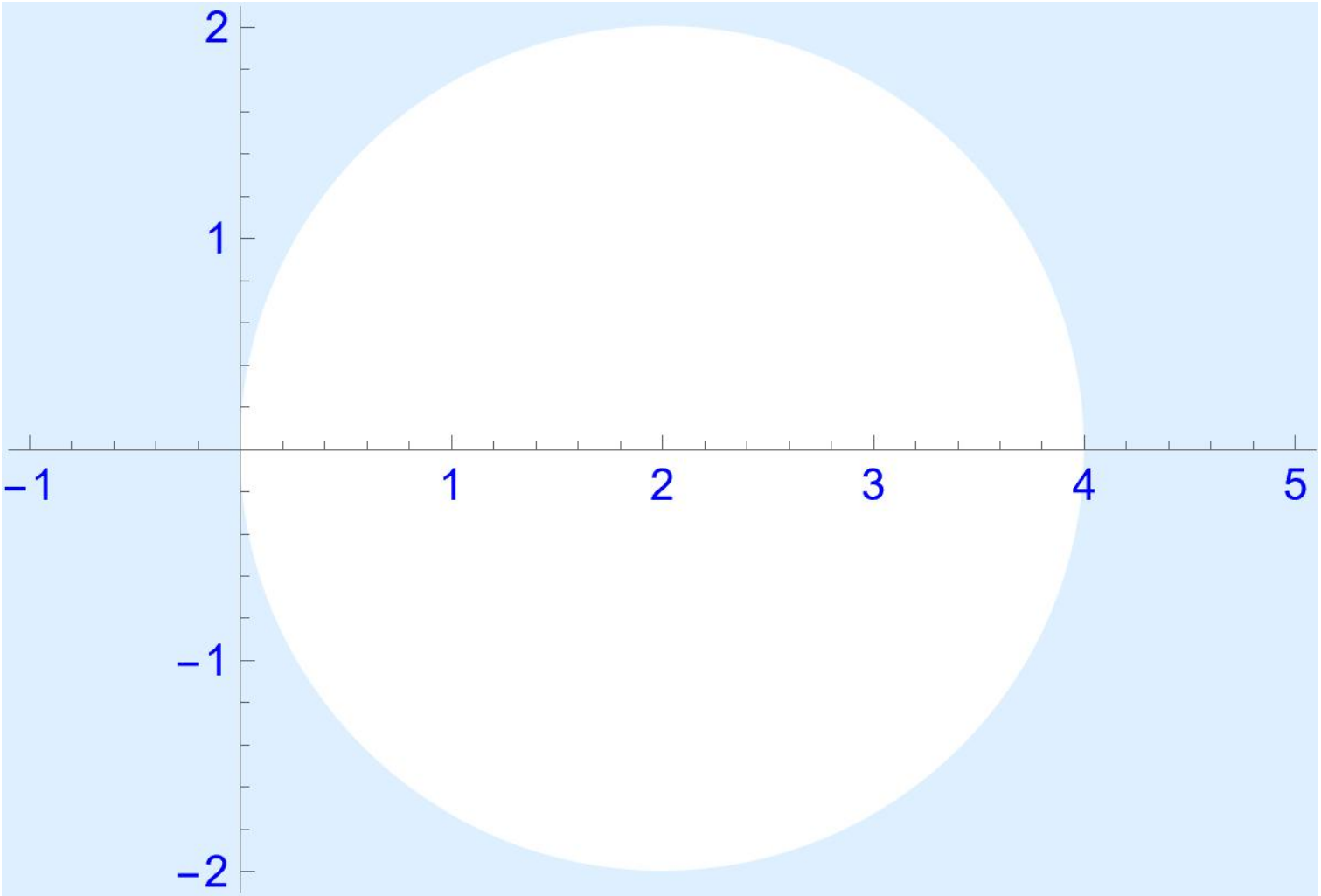
Example 4. Part 2.

First we observe that, similarly to the previous example, $w(z)$ maps a circle $\{z : |z-1| = 2\}$ onto a circle (not a straight line).
Let us find this circle: The function $w = \dfrac{z+1}{z-2}$ maps real line onto real line. An angle between the circle $\{z : |z-1| = 2\}$ and the real line at the points $z_1 = -1$ and $z_2 = 3$ is $\pi/2$. Therefore, the angle between the image of E and the real line at points $w(z_1) = 0$ and $w(z_2) = 4$ must be also $\pi/2$ (conformal mapping). Therefore, an interval $[0, 4]$ must be a diameter of the image circle. Then, the center of the image circle is at the point 2 and its radius is 2.
Therefore, $w(z)$ maps the circle $\{z : |z-1| = 2\}$ onto a circle $|w-2| = 2$.

However, in this case, $w(z)$ maps the disk $E = \{z : |z-1| < 2\}$ onto exterior of the circle $|w-2| = 2$, as it maps a point of the disk $z = -1$ to a point $w(-1) = -2$ in the exterior of the circle $|w-2| = 2$.

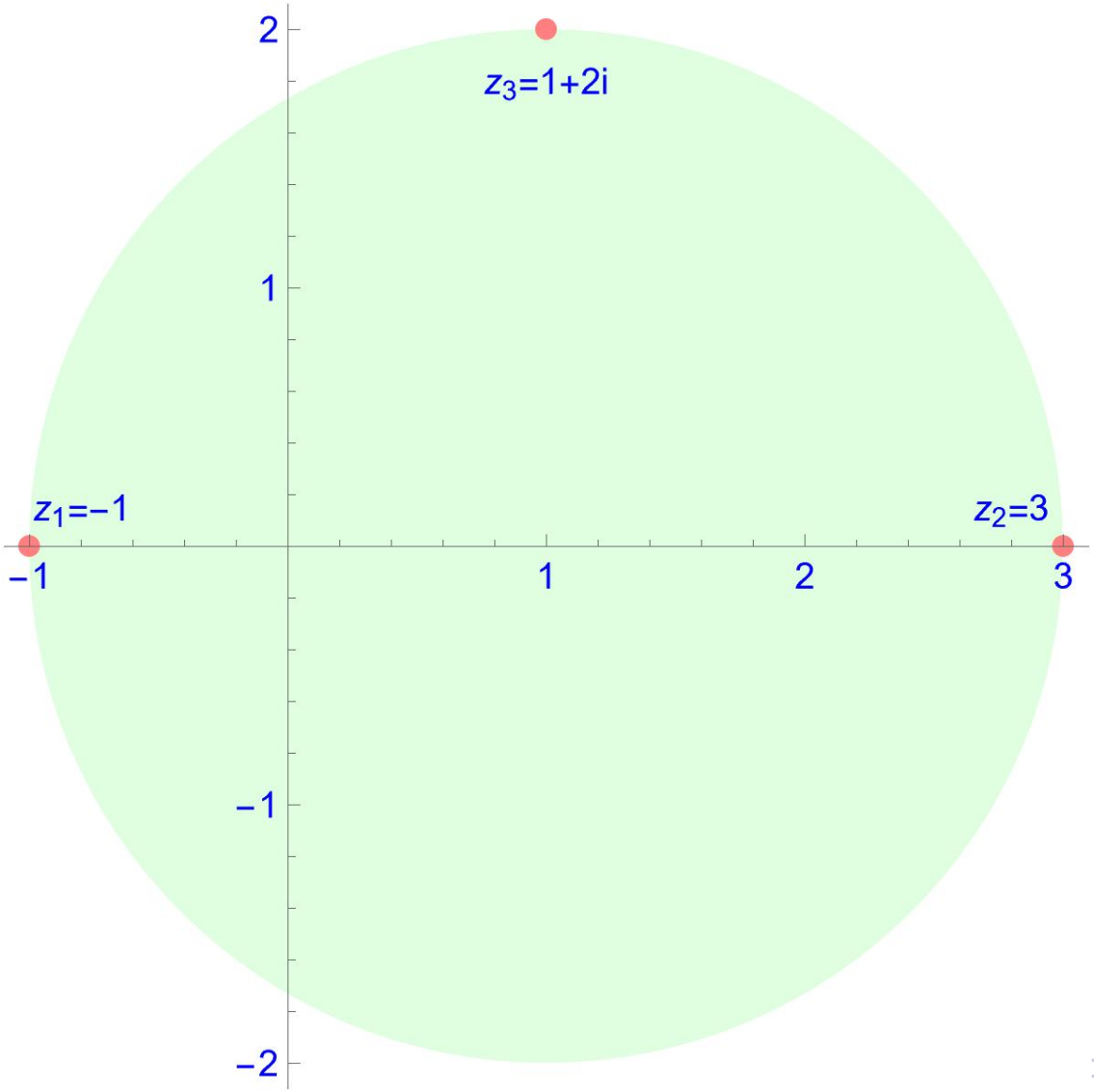
Example 4. Part 3.

Our answer the image of the set $E = \{z : |z-1| < 2\}$ under the map $w = \dfrac{z+1}{z-2}$ is the exterior of the circle $|w-2| = 2$, $w(E) = \{w : |w-2| > 2\}$.



Example 5. Part 1.

Find image of the set $E = \{z : |z-1| < 2\}$ under the map $w = \dfrac{z-1}{2z-6}$



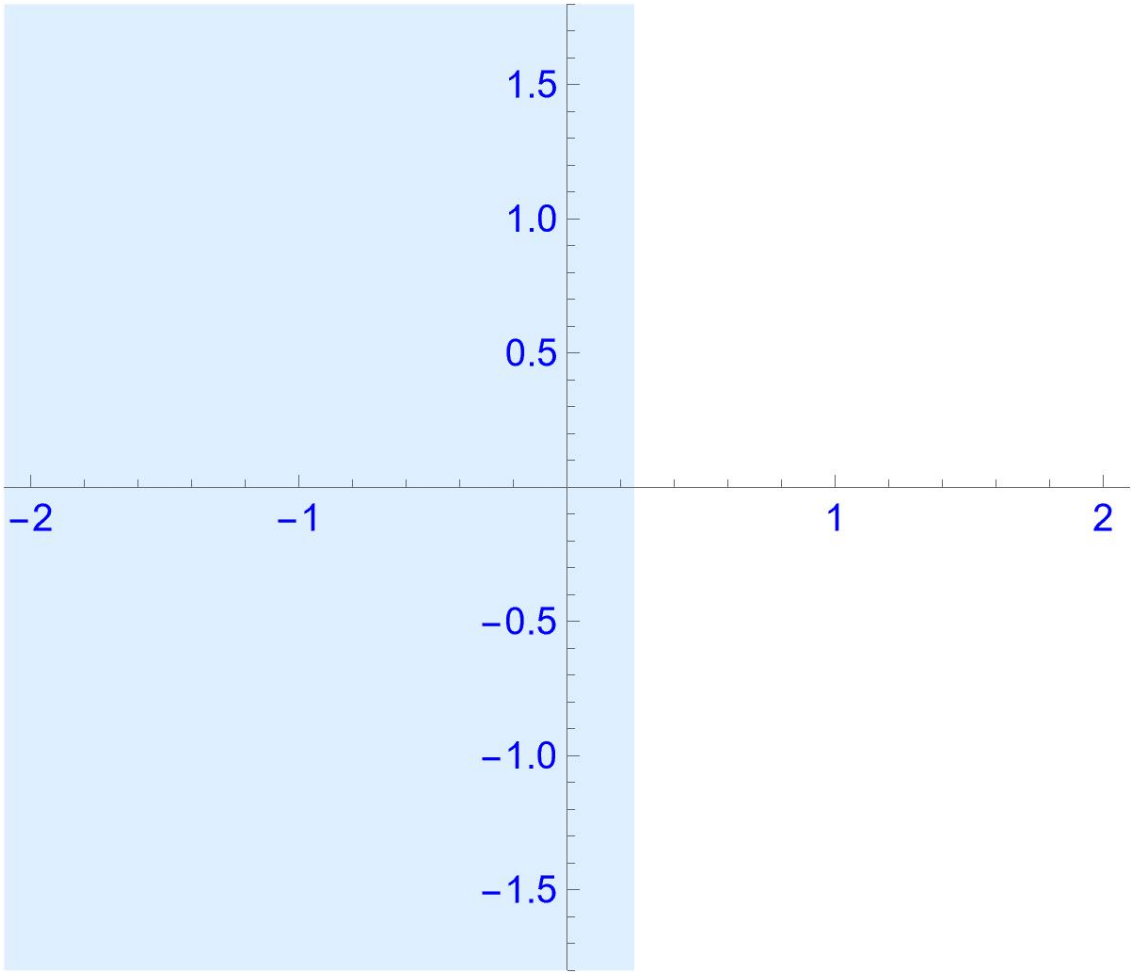
Example 5. Part 2.

In this case, $w(z_2) = w(3) = \infty$, therefore $w(z)$ maps a circle $\{z : |z-1| = 2\}$ onto a straight line.
Let us find this straight line: As in the previous example, our function $w = \dfrac{z-1}{2z-6}$ maps real line onto real line. Again, we will use the facts that the map is conformal and the angle between the circle $\{z : |z-1| = 2\}$ and the real line at the points $z_1 = -1$ is $\pi/2$. Therefore, the angle between the image of E and the real line at point $w(z_1) = 1/4$ must be also $\pi/2$. Therefore, $w(z)$ maps the circle $\{z : |z-1| = 2\}$ onto a vertical line $Re\, w = 1/4$.

Now, we need to find onto which halfplane $w(z)$ map the disc $\{z : |z-1| < 2\}$. Let us pick a point from the interior of the disc E, $z = 1$, and find $w(1)$, which should be in the interior of the image of the disc. $w(1) = 0$ lies to the left of the line $Re\, z = 1/4$. Therefore, $w(E) = \{w : Re\, w < 1/4\}$.

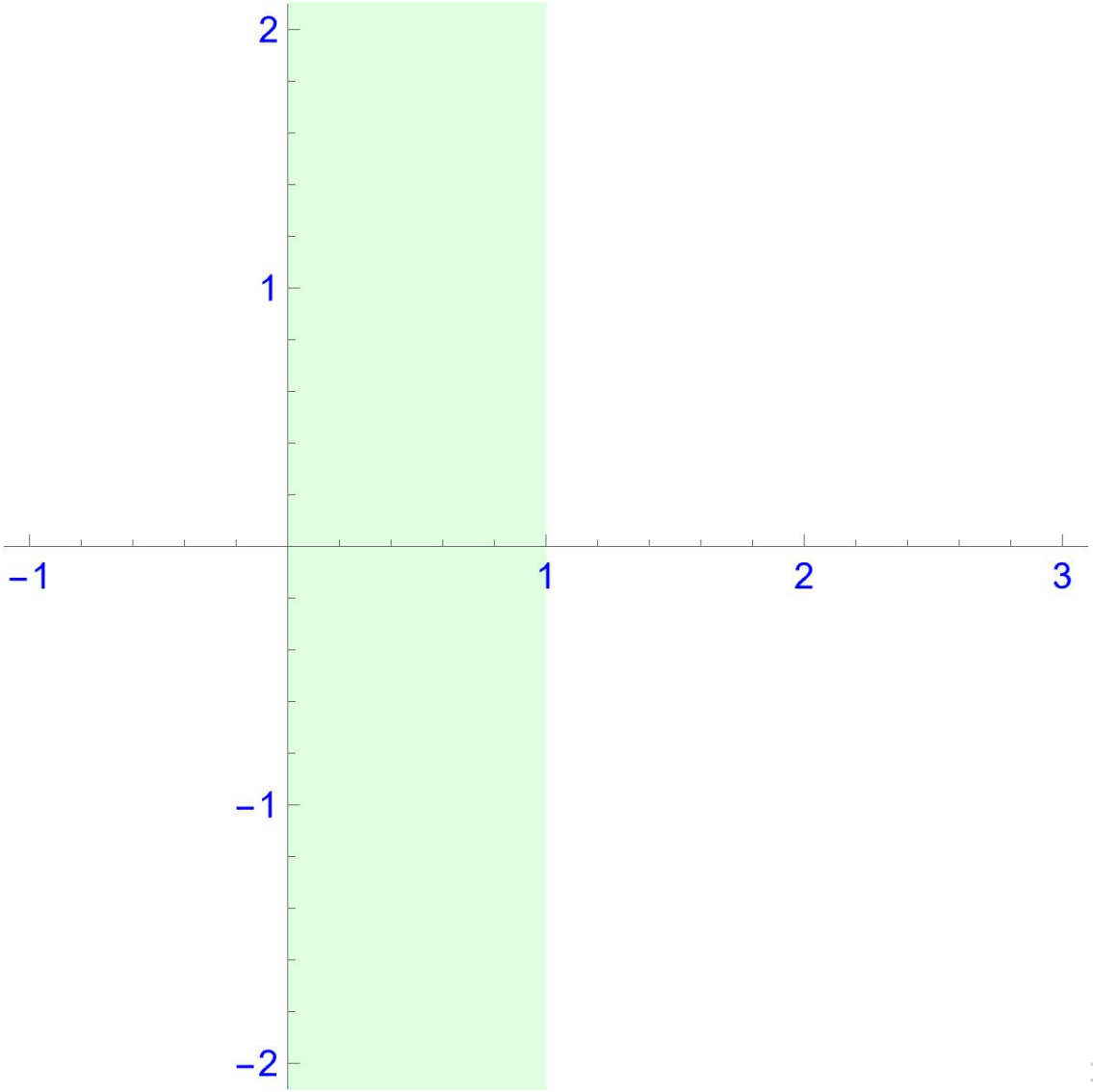
Example 5. Part 3.

Our answer the image of the set $E = \{z : |z-1| < 2\}$ under the map $w = \dfrac{z-1}{2z-6}$ is half-plane $w(E) = \{w : Re\, w < 1/4\}$.



Example 6. Part 1.

Find image of the set $E = \{z : 0 < Re\, z < 1\}$ under the map $w = \dfrac{z-1}{z}$



Example 6. Part 2.

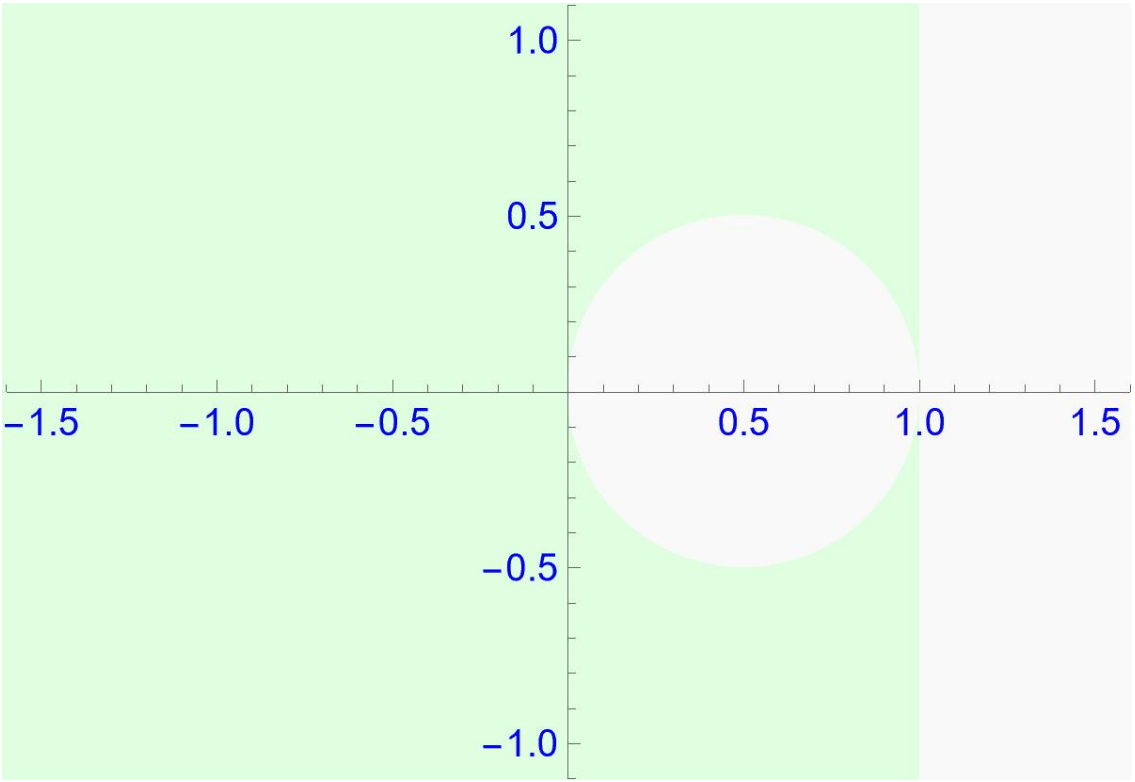
First of all, let us notice that $w(z)$ maps real line onto real line. Then the line $Re\, z = 0$ maps on the straight line, not a circle, as $w(z)$ maps 0 to ∞ and $w(\infty) = 1$. $w(z)$ is conformal at ∞ and, therefore, the vertical line $Re\, w = 1$ is an image of the line $Re\, z = 0$ under this map.

$w(z)$ maps the second line $Re\, z = 1$ onto the circle passing through points 0 and 1. The map $w(z)$ is conformal at both 1 and ∞, therefore the circle is $|w - 1/2| = 1/2$.

Interior point of E, $z = 1/2$, goes into the point $w(1/2) = -1$ outside of the circle. Therefore, the image of E under the $w(z)$ is $w(E) = \{w : Re\, w < 1, |w - 1/2| > 1/2\}$.

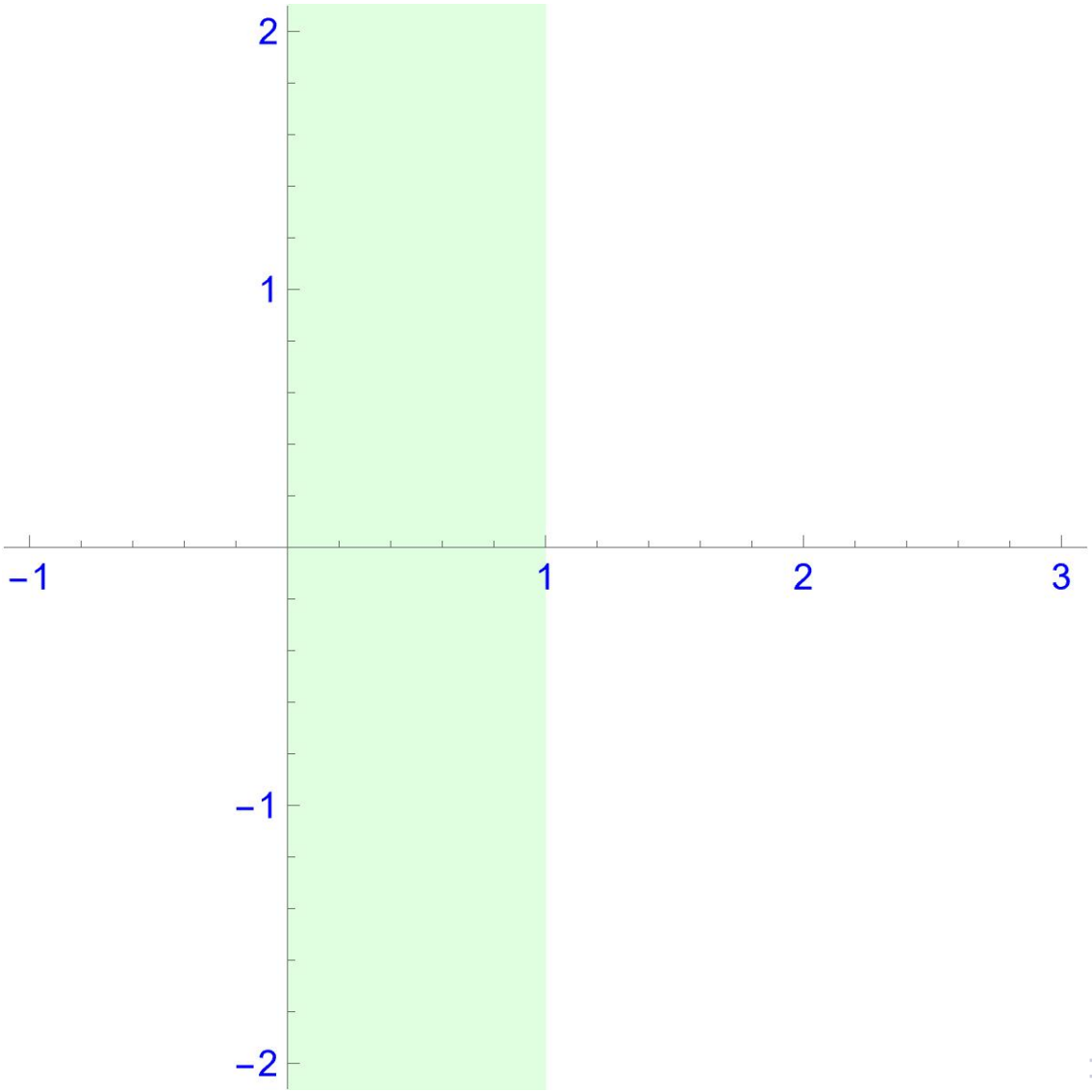
Example 6. Part 3.

Answer: the image of the set $E = \{z : 0 < Re\, z < 1\}$ under the map $w = \dfrac{z-1}{z}$ is $w(E) = \{w : Re\, w < 1, |w - 1/2| > 1/2\}$.



Example 7. Part 1.

Find image of the set $E = \{z : 0 < Re\, z < 1\}$ under the map $w = \dfrac{z-1}{z-2}$



Example 7. Part 2.

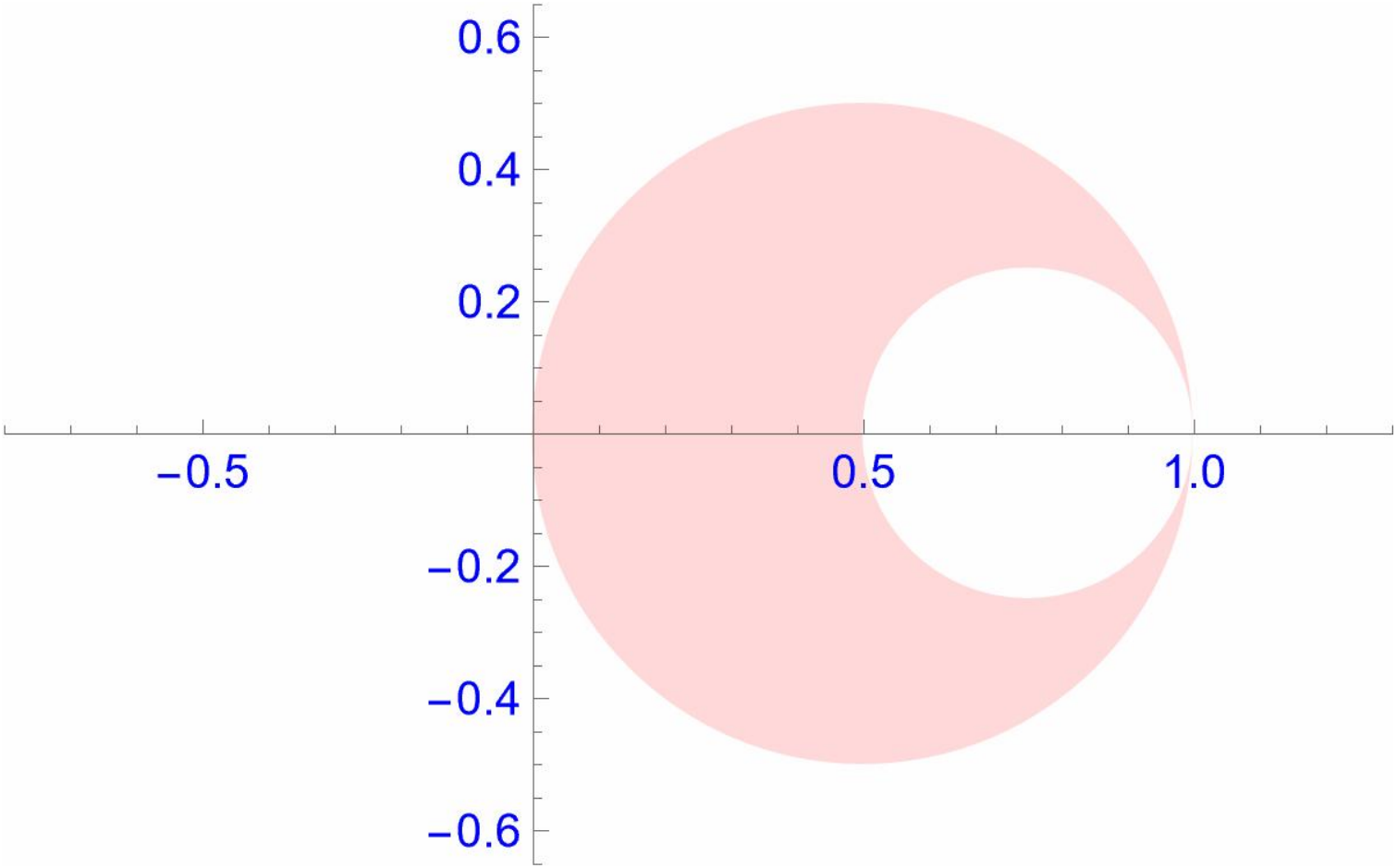
w(z) maps real line onto real line. The images of both lines $Re\, z = 0$ and $Re\, z = 1$ are circles (not straight lines) under $w(z)$. In addition both circles pass through the point $w(\infty) = 1$.

Let us find image of $Re\, z = 0$: we have $w(0) = 1/2$ and $w(\infty) = 1$, and we know that $w(z)$ is a conformal map at both points. Therefore, the angle between real line and the circle at images of both points in w is $\pi/2$. It means that $[1/2, 1]$is the diameter of the circle. Then the equation of this circle is $|w - 3/4| = 1/4$. Similarly we find that the equation of the second circle (image of the line $Re\, z = 1$) is $|w - 1/2| = 1/2$.

Finally, we should see where $w(z)$ maps an interior point of E: let us pick $z = 1/2$, then $w(1/2) = 1/3$. Therefore, $w(E)$ is an area between circles: $w(E) = \{w : |w - 3/4| > 1/4, |w - 1/2| < 1/2\}$.

Example 7. Part 3.

Answer: The image of the set $E = \{z : 0 < Re\, z < 1\}$ under the map $w = \dfrac{z-1}{z-2}$ is $w(E) = \{w : |w - 3/4| > 1/4,\ |w - 1/2| < 1/2\}$.



Thank you